AGASTYA KATARIA

agastyakataria17@gmail.com | Dublin, Ireland | GitHub: github.com/agastya-kataria |
LinkedIn: www.linkedin.com/in/agastyakataria176

PROFILE SUMMARY

Computer Science undergraduate with International Mathematical Olympiad achievements and hands-on experience in quantitative portfolio optimization, risk modelling, and financial analytics.

EDUCATION

B.A. in Computer Science, Trinity College Dublin — First Class Honours (77%) | Sep 2024 – Present
Diploma in Python Programming, EBVTR — A Grade | Dec 2022 – Feb 2023
SAT — 1440 (95th percentile overall, 99th percentile Mathematics) | Feb 2023

KEY SKILLS

Python, Java, Pandas, NumPy, SciPy, Plotly, Matplotlib, PostgreSQL, Flask, FastAPI, React, FIX Protocol, Docker, Kubernetes, Kafka, Prometheus, Financial Modeling, Portfolio Optimization, Risk Management, VaR, Monte Carlo Simulation, Black–Scholes, Duration, Convexity, Yield Curves, Machine Learning (LSTM, PPO), Data Structures & Algorithms, API Development, Data Visualization, Quantitative Research, Asset Allocation, Private Markets Analysis

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PROJECTS

QuantLab-Suite (Creator & Lead Developer) | Mar 2024 – Aug 2025
Institutional-grade multi-asset trading & risk platform with FIX protocol simulation and advanced order types. Built real-time risk engine (VaR: Historical & Monte Carlo), GARCH volatility, and stress testing. Integrated ML (LSTM, PPO) and HFT simulations; optimized latency and throughput. Microservices: Docker, PostgreSQL, Kafka, Prometheus. MiFID II reporting & best execution.

Mini Hedge Fund (AI-Driven) | Aug 2025 – Present
Automated pipeline ingesting Reuters/Bloomberg feeds → FinBERT sentiment → statistical arbitrage & momentum strategies. Full-stack system with spaCy/BERT NLP, Python execution engine, and containerized deployment (Docker/Kubernetes).

Portfolio Management Platform (Creator & Lead Developer) | Feb 2025 – Aug 2025
Python-based portfolio & risk analytics tool implementing MPT, Sharpe ratio maximization, and Monte Carlo scenario

simulations. Implemented Black–Scholes pricing with Greeks, fixed income models (duration, convexity, yield curve valuation), and VaR. Simulated & optimized 1,000+ portfolios; achieved ~25% faster computation via NumPy vectorization; designed Plotly analytics dashboard.

Private Markets Analyzer | Jul 2024 – Present
Analytics suite for private equity, VC, and real assets: ROI, IRR, DPI, RVPI, TVPI calculations; Monte Carlo exit modeling; API-driven (Flask/FastAPI) backend and React front-end.

COURSES & CERTIFICATIONS

Mathematics for Machine Learning (Imperial, Coursera) | Data Structures & Algorithms (UCSD, Coursera) | Finance & Quantitative Modeling (Wharton Online) | Cybersecurity Fundamentals (IBM) | Introduction to CS & Programming using Python (MITx) | Financial Markets (Yale) | Web Dev & API Specialization (Coursera)

ACHIEVEMENTS

• International Mathematical Olympiad — Multiple medals; highest International Rank 13 (2015), other ranks 39 (2016), 52 (2020). Ranked in top 0.05% globally (est.).
• International Science Olympiad — Rank 27.
• Eurofest 2019 — 2nd place, Odyssey of the Mind.

PERSONAL SKILLS

• Willingness to Learn Quickly — Rapidly learn and apply complex concepts, tools, and technologies.
• Networking — Build strong professional relationships through clear communication.
• Time Management — Focused under pressure, manage priorities, meet deadlines.
• Highly Ambitious — Relentless drive to succeed in competitive environments.

LEADERSHIP & ACTIVITIES

• Member, Trinity Student Management Fund (AUM ~€700k)
• Volunteer, Youth Red Cross (District Level Camps)
• STEM Educator, Spread Smile Foundation (Academic mentoring)